



BITS Pilani

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AIML for Robotics

Module 1 & 2

Introduction & Components of Programmable Robotic
System

Section Faculty:

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The course owner and the authors of this course deck, Prof. Raja vadhana and Dr.Sangeetha, are gratefully acknowledging all authors who made their course materials freely available online.

☐ Introduction

- ☐ Types of Robots
- ☐ Mobile Robots – Use case of Programmable solutions

☐ Components of Programmable Robotic System

- ☐ Perception
- ☐ Representation
- ☐ Motion

☐ Localization

- ☐ Taxonomy
- ☐ Types

☐ Mapping

- ☐ Feature Based
- ☐ Grid Based

☐ Path Planning

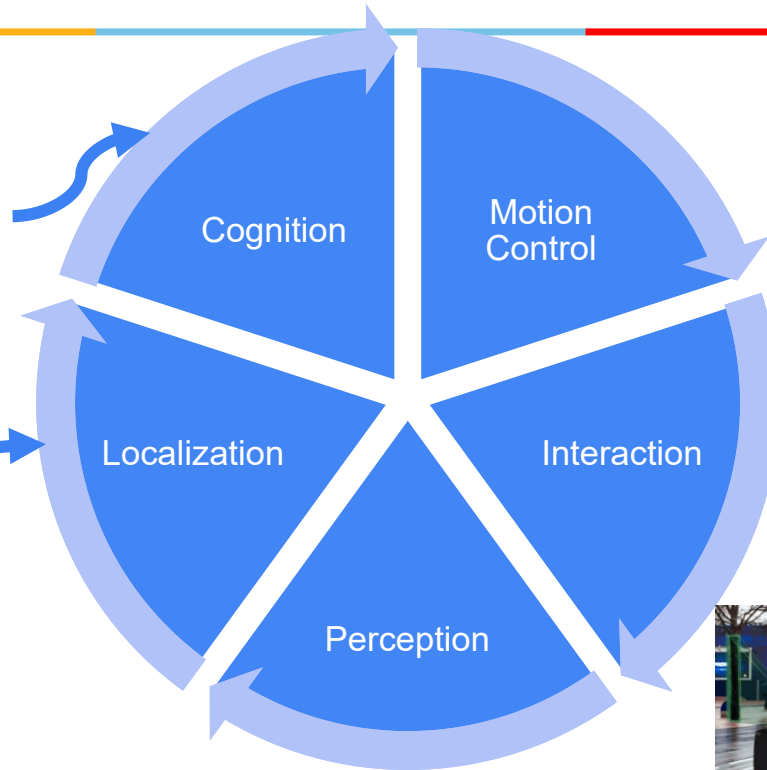
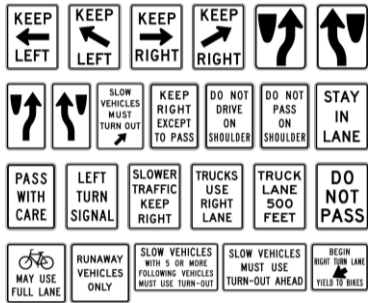
- ☐ Graph Based
- ☐ Sample Based
- ☐ Evolutional Techniques

☐ Human – Robot Interaction

- ☐ Ethics
- ☐ Introduction to Visual Bot
- ☐ Introduction to Chat Bot

Components of Probabilistic Robotics

Perception & Control



Perception & Control



- Perception is the process by which the robot uses its sensors to obtain information about the state of its environment. For example, a robot might take a camera image, a range scan, or query its tactile sensors to receive information about the state of the environment. called a measurement, although we will sometimes also call it observation or percept.
- Control actions change the state of the world. They do so by actively asserting forces on the robot's environment. Examples of control actions include robot motion and the manipulation of objects, a robot may keep a record of all past sensor measurements and control actions. We will refer to such a the collection as the data

Representation

1. State
 1. x_t : [Location of Pedestrian, Location of Building, Pose, Velocity, Battery Status, Battery Energy Level,....]

2. Mobile Robot :
 1. Kinematic State
 2. Dynamic State
 2. [Co-ord x, y, z , Euler Angle θ_1 , θ_2 , θ_3]
(Co-ord x, y, Heading Direction θ) = (14.12,12.7,0.755)
 1. Actuator Config : (JointID, Degree of Freedom....)
 2. Velocity: [v_x , v_y , v_z , v_{θ_1} , v_{θ_2} , v_{θ_3}]

3. Environment:
Object Location,
Features etc.,
 3. [ObjID1, Loc1, Loc2, Feature1, Feature2, Velocities, ObjID2.....]

Note : The three angular orientation is also called euler's angle named "Pitch", "Roll", "Yaw". For planar env only three variable is enough to represent the robot pose instead of 6.

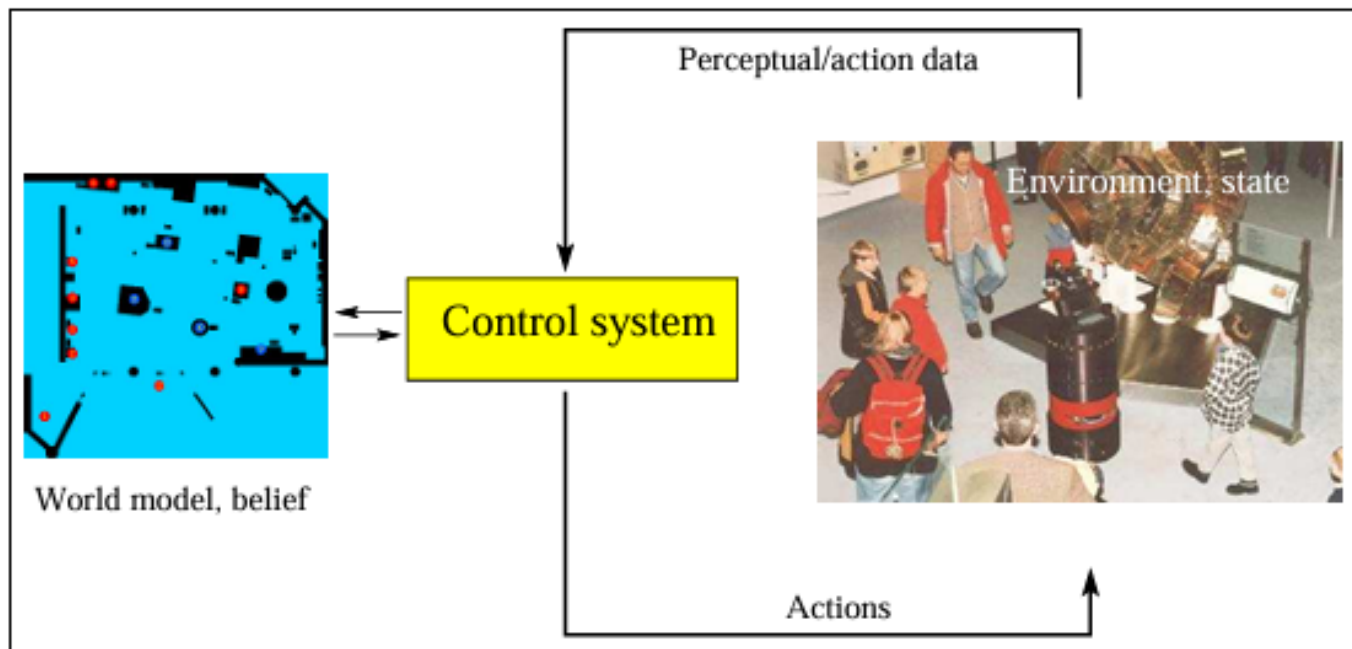
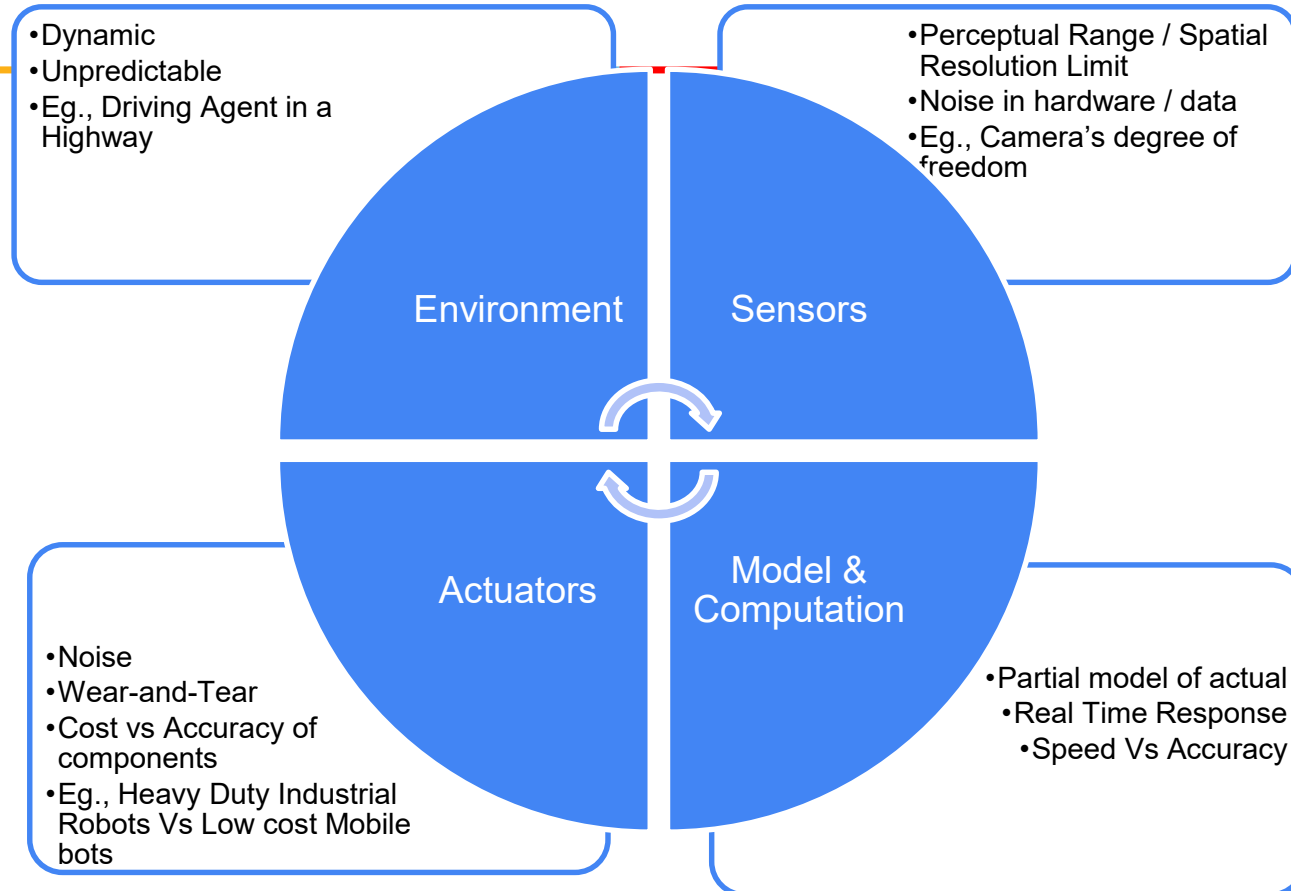


Figure 2.1 Robot Environment Interaction.

Uncertainty



Estimation of State

1. Percept Measurement Data - Momentary State of the Env

2. Motion Control Data - Change of State in Env

$$1. z_{t1:t2} = z_{t1}, z_{t1+1}, z_{t1+2}, \dots, z_{t2}$$

$$2. u_{t1:t2} = u_{t1}, u_{t1+1}, u_{t1+2}, \dots, u_{t2}$$

3. Prediction : Control Update : $p(x_t | z_{1:t-1}, u_{1:t})$:
Control $\rightarrow$ Infer $\rightarrow$ Perceive

Correction : $p(x_t | z_{1:t}, u_{1:t})$: Control $\rightarrow$ Perceive $\rightarrow$ Infer
(Measurement Update) : Predict $\rightarrow$ Infer

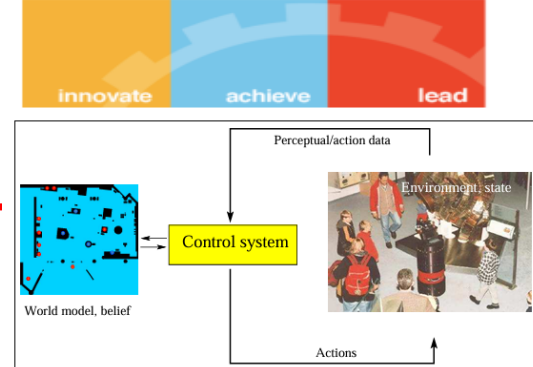


Figure 2.1 Robot Environment Interaction.

X_0	$P(X_0)$
is_open	0.5
is_closed	0.5

X_{t-1}	U_t	$P(X_t = \text{is_open})$	$P(X_t = \text{is_closed})$
is_open	push	1	0
is_closed	Push	0.8	0.2
is_open	do_nothing	1	0
is_closed	do_nothing	0	1

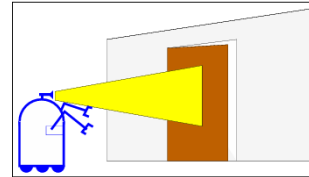
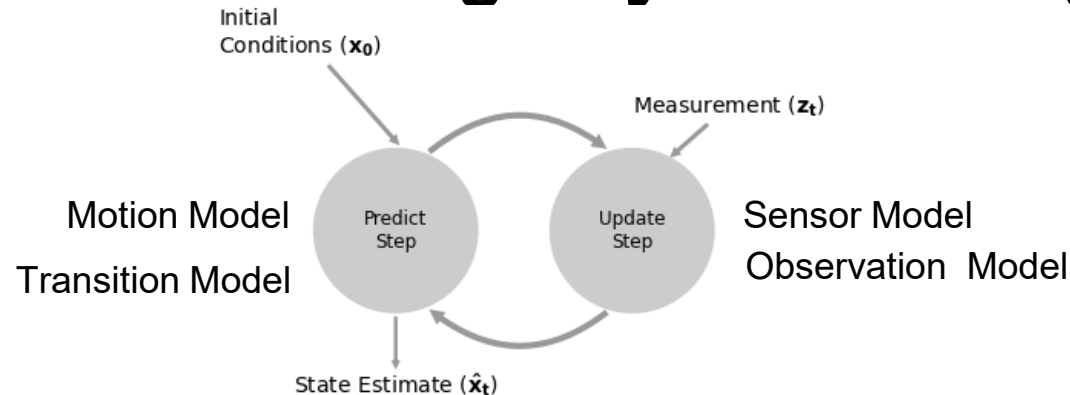


Figure 2.2 A mobile robot estimating the state of a door.

X_t	$P(Z_t = \text{sense_open})$	$P(Z_t = \text{sense_closed})$
is_open	0.6	0.4
is_closed	0.2	0.8

State Estimation using Bayes Filter Algorithm



Recursive State Estimation

Bayes Filter Algorithm

1. Initial Belief
2. Control Update
3. Measurement Update

X_{t-1}	U_t	$P(X_t = \text{is_open})$	$P(X_t = \text{is_closed})$
is_open	push	1	0
is_closed	Push	0.8	0.2
is_open	do_nothing	1	0
is_closed	do_nothing	0	1



X_0	$P(X_0)$
is_open	0.5
is_closed	0.5

X_t	$P(Z_t = \text{sense_open})$	$P(Z_t = \text{sense_closed})$
is_open	0.6	0.4
is_closed	0.2	0.8

Predict the state of the door, given robot's action & sensor data observed :

(?, do_nothing, sense_open, push, sense_open)
(?, A1, S2, A2, S3)

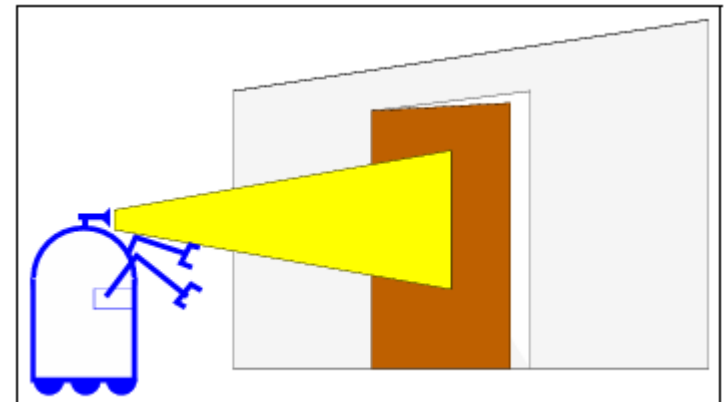


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Hidden Markov Model



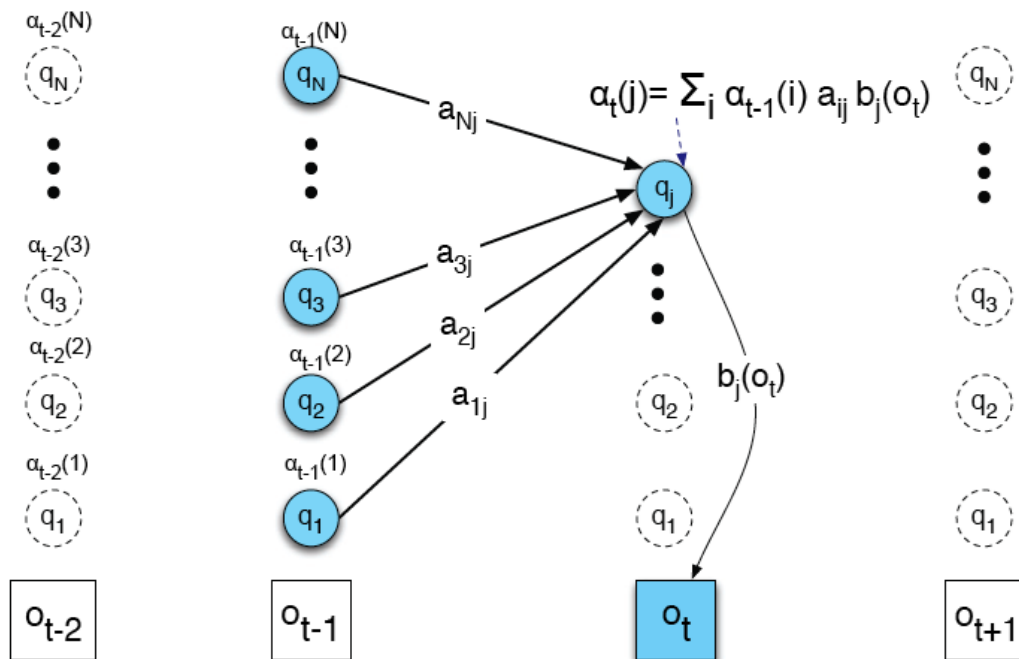
Filtering

$$P(X_2 | \text{Open, Closed})$$

$$P(X_t | E_{1...t})$$

In your Text book another example for these inferences is explained
“Task of predicting the weather condition by a security personnel
sitting in an underground secret installation by
observing the state of an employee who either umbrella or don’t”.
Kindly check it and work it out as additional practice

Observation Likelihood – Forward Propagation



Source Credit : Speech and Language Processing - Jurafsky and Martin

Observation Likelihood – Forward Propagation

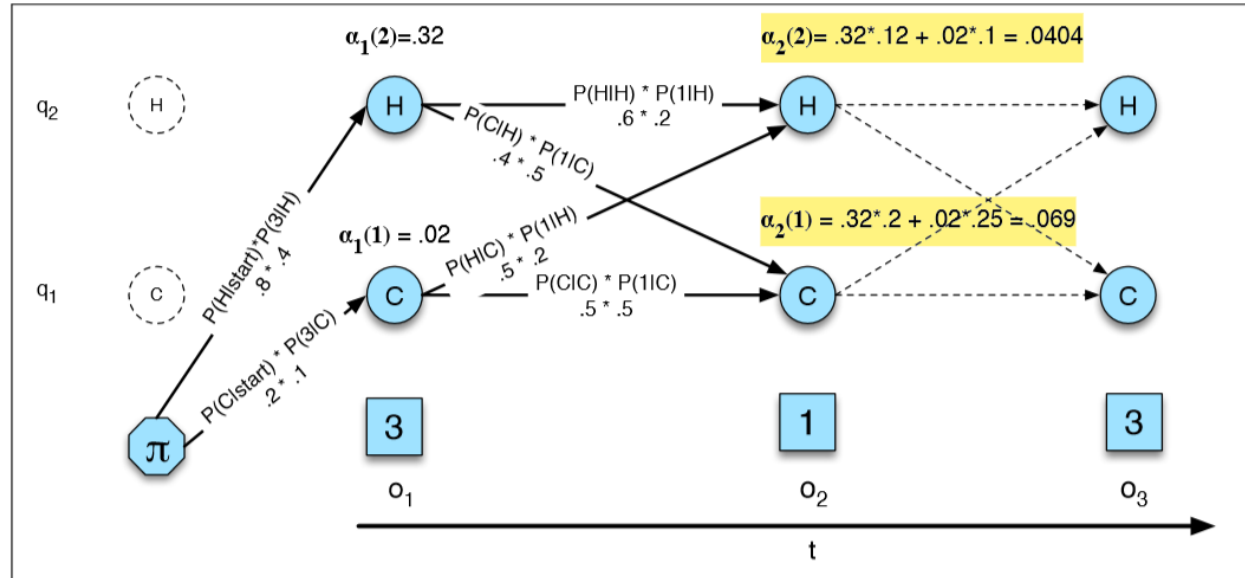
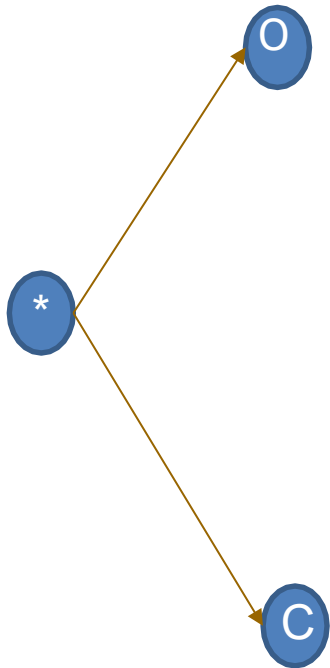


Figure A.5 The forward trellis for computing the total observation likelihood for the ice-cream events 3 1 3. Hidden states are in circles, observations in squares. The figure shows the computation of $\alpha_t(j)$ for two states at two time steps. The computation in each cell follows Eq. A.12: $\alpha_t(j) = \sum_{i=1}^N \alpha_{t-1}(i) a_{ij} b_j(o_t)$. The resulting probability expressed in each cell is Eq. A.11: $\alpha_t(j) = P(o_1, o_2 \dots o_t, q_t = j | \lambda)$.

Source Credit : Speech and Language Processing - Jurafsky and Martin

Recursive State Estimation

Bayes Filter Algorithm



X_{t-1}	U_t	$P(X_t = \text{is_open})$	$P(X_t = \text{is_closed})$
is_open	push	1	0
is_closed	Push	0.8	0.2
is_open	do_nothing	1	0
is_closed	do_nothing	0	1

X_0	$P(X_0)$
is_open	0.5
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X_t	$P(Z_t = \text{sense_open})$	$P(Z_t = \text{sense_closed})$
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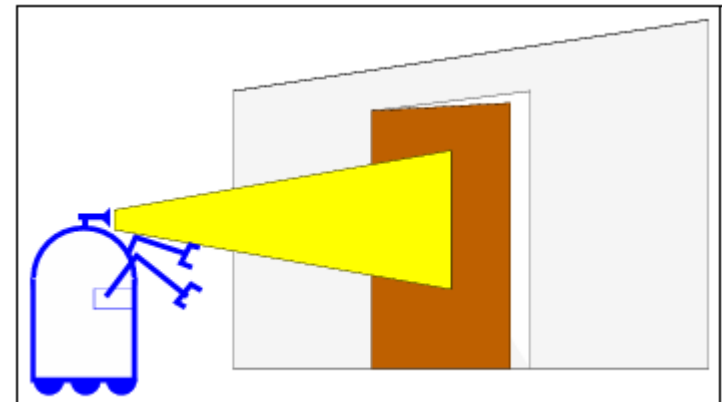


Figure 2.2 A mobile robot estimating the state of a door.

Recursive State Estimation

Bayes Filter Algorithm

1. Initial Belief
2. Control Update
3. Measurement Update

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is_open	0.5
is_closed	0.5

X_t	$P(Z_t = \text{sense_open})$	$P(Z_t = \text{sense_closed})$
is_open	0.6	0.4
is_closed	0.2	0.8

$$\begin{aligned}
 &P(X_{1=\text{is_open}}) \\
 &= p(X_{1=\text{is_open}} | U_{1=\text{do_nothing}}, X_{0=\text{is_open}}) * P(X_{0=\text{is_open}}) \\
 &+ \\
 &p(X_{1=\text{is_open}} | U_{1=\text{do_nothing}}, X_{0=\text{is_closed}}) * P(X_{0=\text{is_closed}}) \\
 &= 1 * 0.5 + 0 * 0.5 \\
 &= 0.5
 \end{aligned}$$

$$\begin{aligned}
 &P(X_{1=\text{is_closed}}) \\
 &= p(X_{1=\text{is_closed}} | U_{1=\text{do_nothing}}, X_{0=\text{is_open}}) * P(X_{0=\text{is_open}}) \\
 &+ \\
 &p(X_{1=\text{is_closed}} | U_{1=\text{do_nothing}}, X_{0=\text{is_closed}}) * P(X_{0=\text{is_closed}}) \\
 &= 0 * 0.5 + 1 * 0.5 \\
 &= 0.5
 \end{aligned}$$

$u_1 = \text{do_nothing}$ and $z_1 = \text{sense_open}$

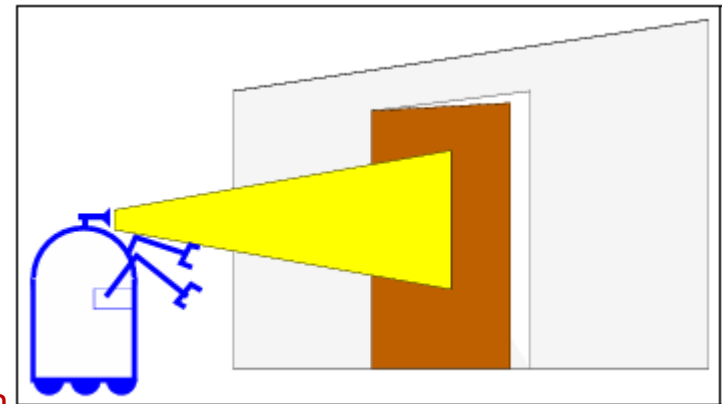


Figure 2.2 A mobile robot estimating the state of a door.

Recursive State Estimation

Bayes Filter Algorithm

1. Initial Belief
2. Control Update
3. Measurement Update

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X_0	$P(X_0)$
is_open	0.5
is_closed	0.5

X_t	$P(Z_t = \text{sense_open})$	$P(Z_t = \text{sense_closed})$
is_open	0.6	0.4
is_closed	0.2	0.8

$$\text{bel}(x_1) = \eta p(Z_1 = \text{sense_open} | x_1) * p(x_1)$$

$$\begin{aligned} \text{bel}(X_{1=\text{is_open}}) &= \eta p(Z_1 = \text{sense_open} | X_{1=\text{is_open}}) * p(X_{1=\text{is_open}}) \\ &= \eta 0.6 * 0.5 \\ &= \eta 0.3 \end{aligned}$$

$$\begin{aligned} \text{bel}(X_{1=\text{is_closed}}) &= \eta p(Z_1 = \text{sense_open} | X_{1=\text{is_closed}}) * \\ &p(X_{1=\text{is_closed}}) \\ &= \eta 0.2 * 0.5 \\ &= \eta 0.1 \end{aligned}$$

$$\begin{aligned} \text{bel}(X_{1=\text{is_open}}) &= 0.75 \\ \text{bel}(X_{1=\text{is_closed}}) &= 0.25 \end{aligned}$$

$u_1 = \text{do_nothing}$ and $z_1 = \text{sense_open}$

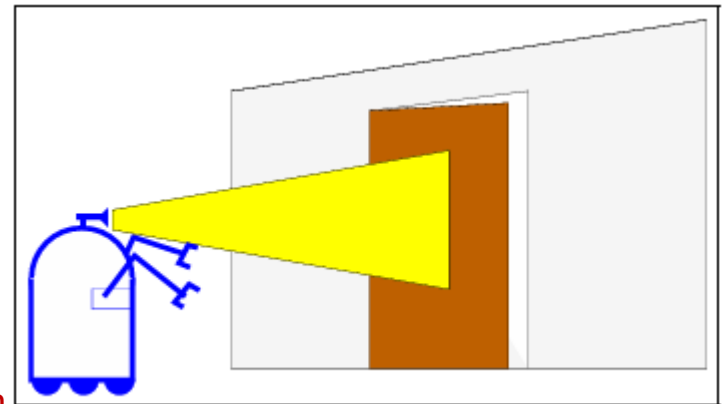
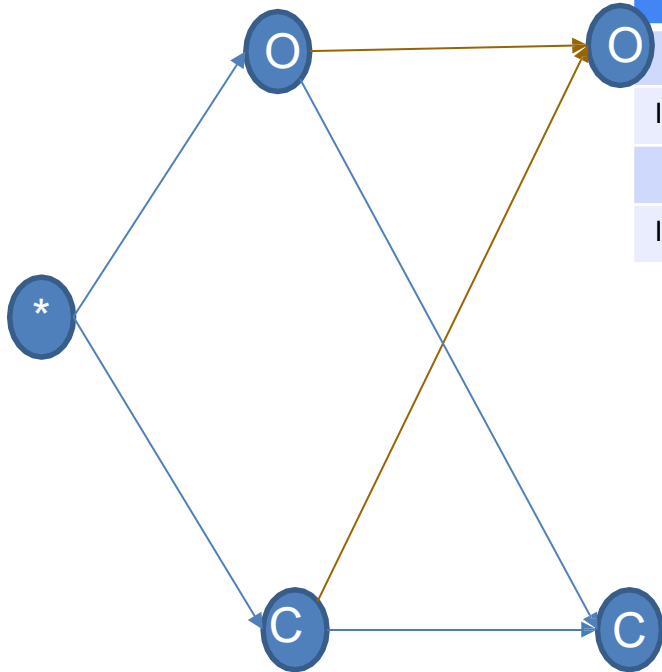


Figure 2.2 A mobile robot estimating the state of a door.

Recursive State Estimation

Bayes Filter Algorithm



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X_0	$P(X_0)$
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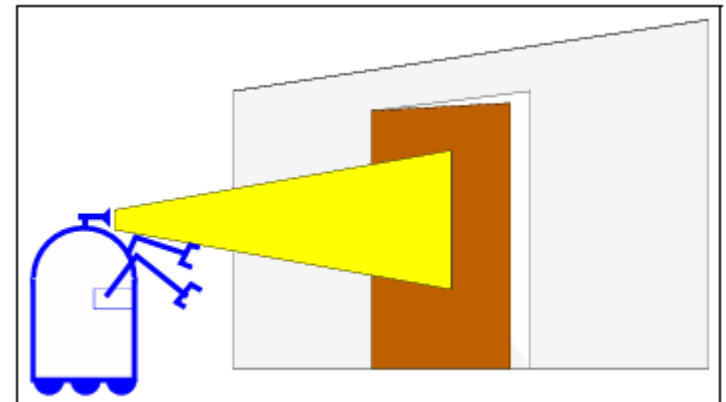


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Recursive State Estimation

Bayes Filter Algorithm

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2. Control Update
3. Measurement Update

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X_0	$P(X_0)$
is_open	0.5
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X_t	$P(Z_t = \text{sense_open})$	$P(Z_t = \text{sense_closed})$
is_open	0.6	0.4
is_closed	0.2	0.8

$u_2 = \text{push}$ and $z_2 = \text{sense_open}$

$$\overline{\text{bel}}(X_2 = \text{is_open}) = 1 \cdot 0.75 + 0.8 \cdot 0.25 = 0.95$$

$$\overline{\text{bel}}(X_2 = \text{is_closed}) = 0 \cdot 0.75 + 0.2 \cdot 0.25 = 0.05$$

and

$$\text{bel}(X_2 = \text{is_open}) = \eta \cdot 0.6 \cdot 0.95 \approx 0.983$$

$$\text{bel}(X_2 = \text{is_closed}) = \eta \cdot 0.2 \cdot 0.05 \approx 0.017$$

Inference :

The robot believes that with 0.983 probability the door is open

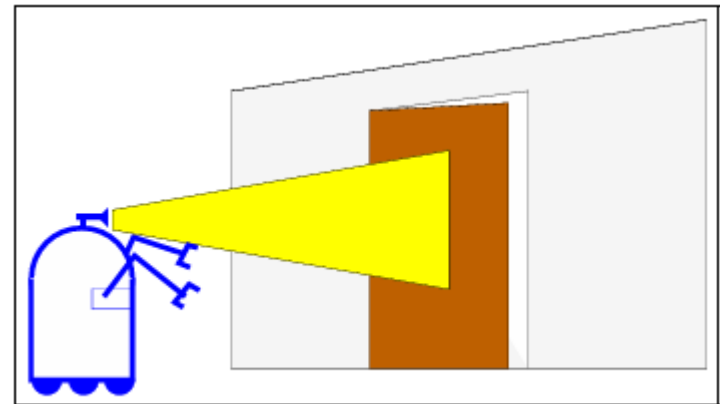


Figure 2.2 A mobile robot estimating the state of a door.

Recap

1. Initial Belief
2. Control Update
3. Measurement Update

$u_2 = \text{push}$ and $z_2 = \text{sense_open}$

MOTION MODEL : $P(X_t | U_t, X_{t-1})$

X_{t-1}	U_t	$P(X_t = \text{is_open})$	$P(X_t = \text{is_closed})$
is_open	push	1	0
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X_0	$P(X_0)$
is_open	0.5
is_closed	0.5

X_t	$P(Z_t = \text{sense_open})$	$P(Z_t = \text{sense_closed})$
is_open	0.6	0.4
is_closed	0.2	0.8

$$\begin{aligned}\overline{bel}(X_2 = \text{is_open}) &= 1 \cdot 0.75 + 0.8 \cdot 0.25 = 0.95 \\ \overline{bel}(X_2 = \text{is_closed}) &= 0 \cdot 0.75 + 0.2 \cdot 0.25 = 0.05\end{aligned}$$

and

$$\begin{aligned}bel(X_2 = \text{is_open}) &= \eta \cdot 0.6 \cdot 0.95 \approx 0.983 \\ bel(X_2 = \text{is_closed}) &= \eta \cdot 0.2 \cdot 0.05 \approx 0.017\end{aligned}$$

Inference :

The robot believes that with 0.983 probability the door is open

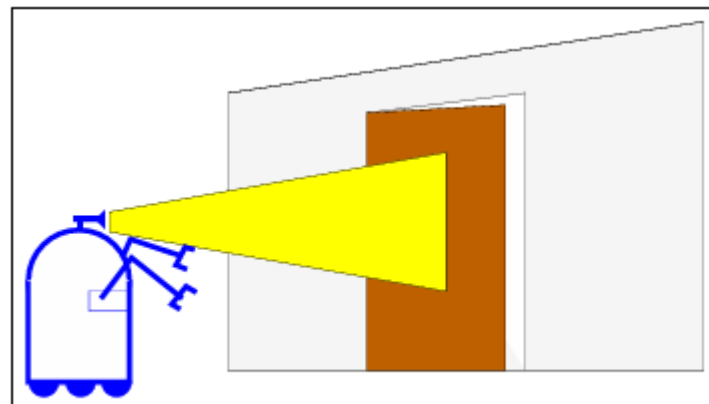


Figure 2.2 A mobile robot estimating the state of a door.

Recap

1. Initial Belief
2. Control Update
3. Measurement Update

$u_2 = \text{push}$ and $z_2 = \text{sense_open}$

$$\overline{\text{bel}}(X_2 = \text{is_open}) = 1 \cdot 0.75 + 0.8 \cdot 0.25 = 0.95$$

$$\overline{\text{bel}}(X_2 = \text{is_closed}) = 0 \cdot 0.75 + 0.2 \cdot 0.25 = 0.05$$

and

$$\text{bel}(X_2 = \text{is_open}) = \eta \cdot 0.6 \cdot 0.95 \approx 0.983$$

$$\text{bel}(X_2 = \text{is_closed}) = \eta \cdot 0.2 \cdot 0.05 \approx 0.017$$

Inference :

The robot believes that with 0.983 probability the door is open

X_{t-1}	U_t	$P(X_t = \text{is_open})$	$P(X_t = \text{is_closed})$
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**SENSOR MEASUREMENT
MODEL : $P(Z_t | X_t)$**

X_t	$P(Z_t = \text{sense_open})$	$P(Z_t = \text{sense_closed})$
is_open	0.6	0.4
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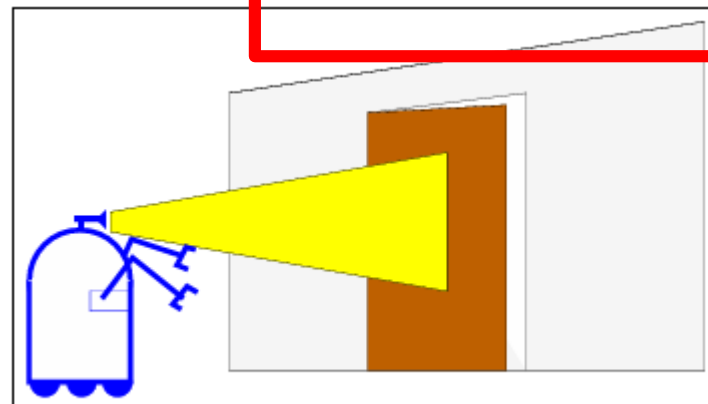


Figure 2.2 A mobile robot estimating the state of a door.

Recap

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3. Measurement Update

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SENSOR MEASUREMENT MODEL : $P(Z_t | X_t)$

X_t	$P(Z_t = \text{sense_open})$	$P(Z_t = \text{sense_closed})$
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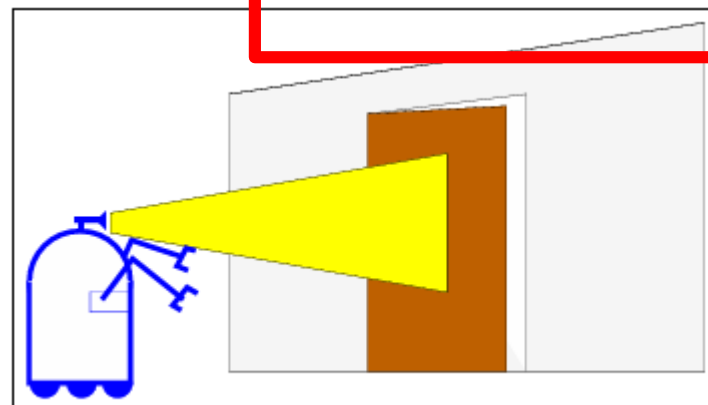
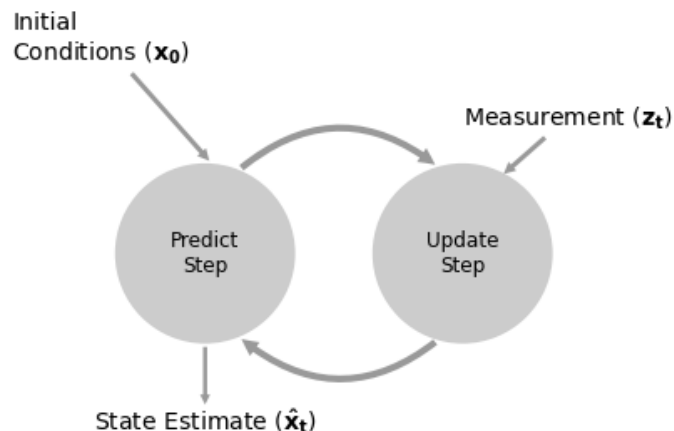


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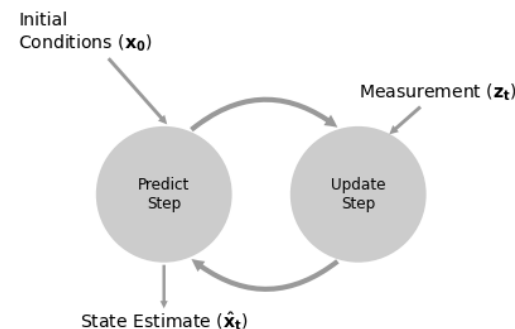
Inference :

The robot believes that with 0.983 probability the door is open

```

1:  Algorithm Discrete_Bayes_filter( $\{p_{k,t-1}\}, u_t, z_t$ ):
2:    for all  $k$  do
3:       $\bar{p}_{k,t} = \sum_i p(X_t = x_k \mid u_t, X_{t-1} = x_i) p_{i,t-1}$ 
4:       $p_{k,t} = \eta p(z_t \mid X_t = x_k) \bar{p}_{k,t}$ 
5:    endfor
6:    return  $\{p_{k,t}\}$ 

```



- Bayes filter cannot be implemented on a digital computer, probabilistic algorithms use tractable approximations for large complex problems. Such approximations may be evaluated according to different criteria, relating to their accuracy, efficiency, and ease of implementation.
- Solution** : Particle Filter , Kalman Filter Approaches to estimate the prior probability

1. Initial Belief
2. Control Update
3. Measurement Update

MOTION MODEL : $P(X_t | U_t, X_{t-1})$

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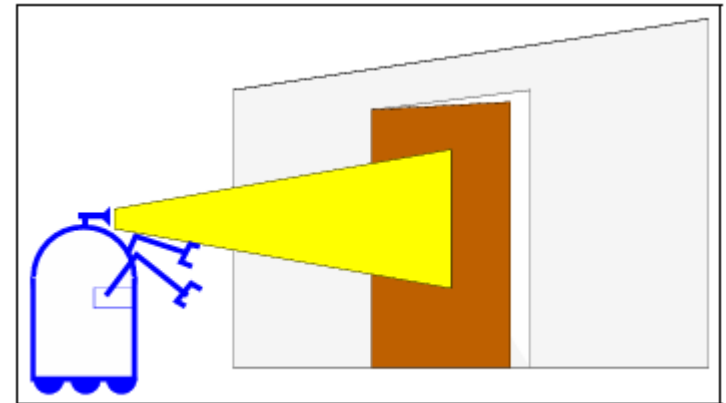
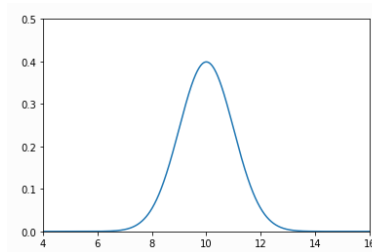


Figure 2.2 A mobile robot estimating the state of a door.



What if the state space is continuous?

Solution : Prefer parametric model like Kalman Filters

Recursive State Estimation

Parametric Model



1. Initial Belief is unimodal Gaussian
2. Transition Probability is a linear function with independent Gaussian noise
3. Measurement Probability is a linear function with Gaussian noise
4. Limited uncertainty

$$\begin{bmatrix} x \\ y \\ \theta \end{bmatrix} = \begin{bmatrix} 50 \\ 15 \\ 10 \end{bmatrix} \quad \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 10 \text{ N} \\ 30^\circ \end{bmatrix}$$

$$\begin{aligned} p(x) &= \frac{1}{\sqrt{2\pi\Sigma}} e^{-\frac{(x-\mu)^2}{2\Sigma}} \\ &= \frac{1}{\sqrt{2\pi\Sigma}} e^{-\frac{(x-\mu)^T \Sigma^{-1} (x-\mu)}{2}} \end{aligned}$$

Use case :

Robot needs to estimate the state in a tracking application. :

- ✓ Real time tracking is not needed.
 - ✓ Computational resources are limited.
- Prior state of the system is known

State Prob:

$$\begin{aligned} p(X_t | U_t, X_{t-1}) &= \frac{1}{\sqrt{2\pi\Sigma}} e^{-\frac{(x-\mu)^2}{2\Sigma}} \\ &= \frac{1}{\sqrt{2\pi R_t}} e^{-\frac{(X_t - (A_t X_{t-1} + B_t U_t))^T R_t^{-1} (X_t - A_t X_{t-1} - B_t U_t)}{2}} \end{aligned}$$

Recursive State Estimation

Parametric Model



1. Kalman Filter

State: $p(X_t | U_t, X_{t-1})$

$$= \frac{1}{\sqrt{2\pi R_t}} e^{-\frac{(X_t - (A_t X_{t-1} + B_t U_t))^T R_t^{-1} (X_t - A_t X_{t-1} - B_t U_t)}{2}}$$

$$\begin{bmatrix} x \\ y \\ \theta \end{bmatrix} = \begin{bmatrix} 50 \\ 15 \\ 10 \end{bmatrix} \quad \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 10 \text{ N} \\ 30^\circ \end{bmatrix}$$

$$\begin{aligned} p(x) &= \frac{1}{\sqrt{2\pi\Sigma}} e^{-\frac{(x-\mu)^2}{2}} \\ &= \frac{1}{\sqrt{2\pi\Sigma}} e^{-\frac{(x-\mu)^T \Sigma^{-1} (x-\mu)}{2}} \end{aligned}$$

Measurement: $p(Z_t | X_t)$

$$= \frac{1}{\sqrt{2\pi Q_t}} e^{-\frac{(Z_t - (C_t X_t))^T Q_t^{-1} (Z_t - C_t X_t)}{2}}$$

Initial Belief: $p(X_0) = \text{bel}(X_0)$

$$= \frac{1}{\sqrt{2\pi\Sigma_0}} e^{-\frac{(X_t - \mu_0)^T \Sigma_0^{-1} (X_t - \mu_0)}{2}}$$

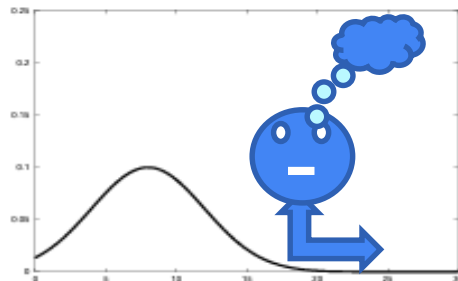
Recursive State Estimation

Parametric Model

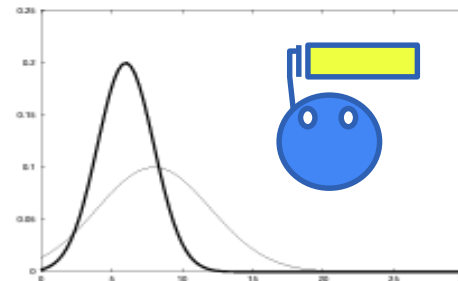
1. Kalman Filter

A robot that moves with constant translational and rotational velocity typically moves on a circular trajectory, which cannot be described by linear next state transitions.

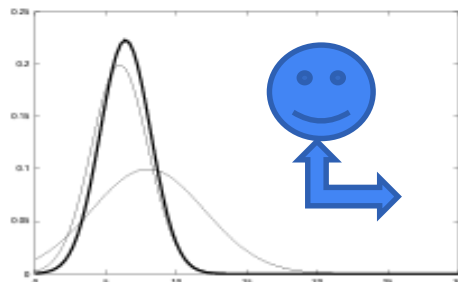
Can the KF be customized for non-linearity?



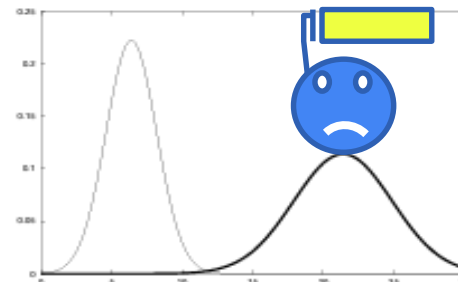
(a)



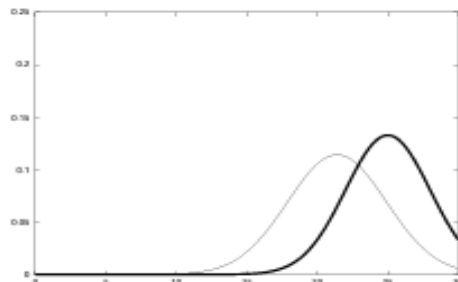
(b)



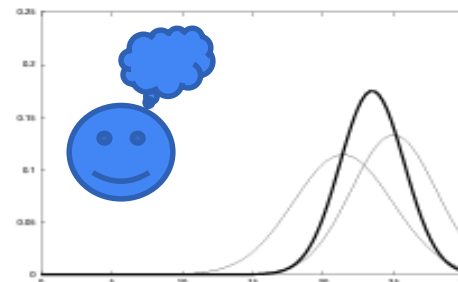
(c)



(d)



(e)



(f)

What if the state space is continuous and there is too much uncertainty to handle by Exact models?

Solution : Prefer non-parametric model like eg., Particle Filters

Recursive State Estimation

Non-Parametric Model



1. Initial Belief is Multimodal and non-Gaussian
2. Too much uncertainty , Multiple Hypothesis exists for decision

Use case :

Robot needs to estimate the state in a tracking application. :

- ✓ Real time tracking is needed.
- ✓ Computational resources are unlimited.
- ✓ Accuracy of methods needs to be adaptive to the real time requirement & available computation.

Idea:

Approximate the posterior by a finite number of parameters.

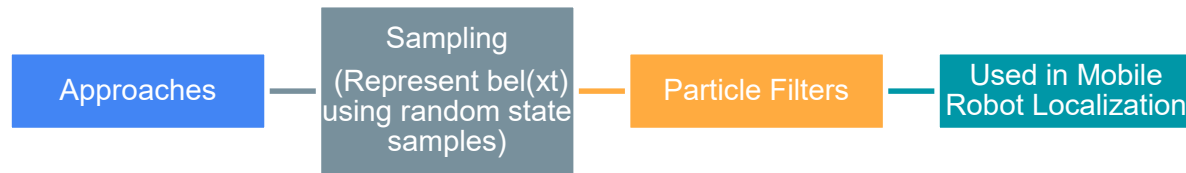
Different methods exist to generate these parameters, and use it to populate the state space

Recursive State Estimation

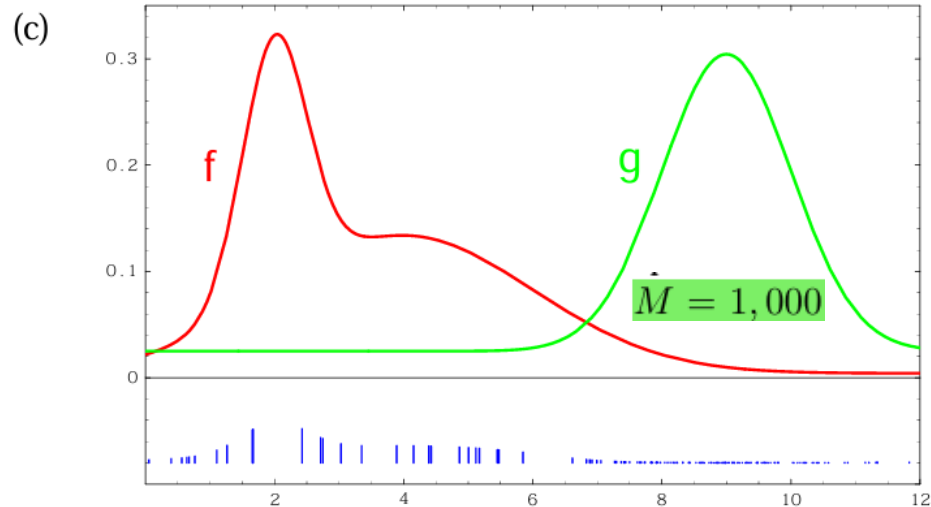
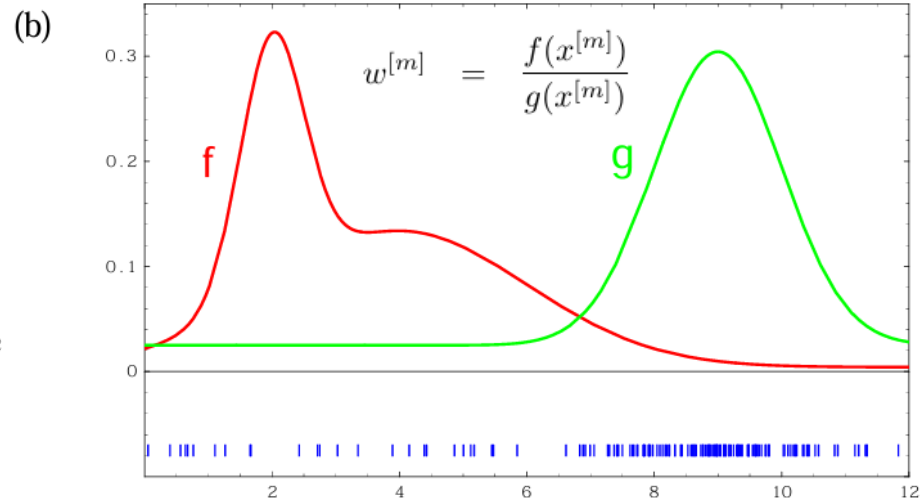
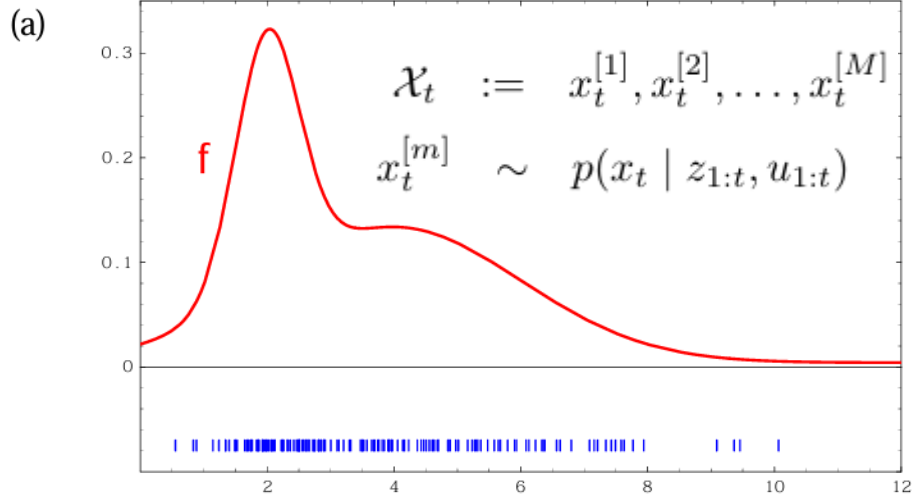
Non-Parametric Model



- Approximate the posterior by a finite number of parameters.
- **A particle set represents a discrete distribution, whereas f is continuous**



- Proposal Distribution (g) vs Target Distribution (f) , Importance Sampling $W(m)$



```

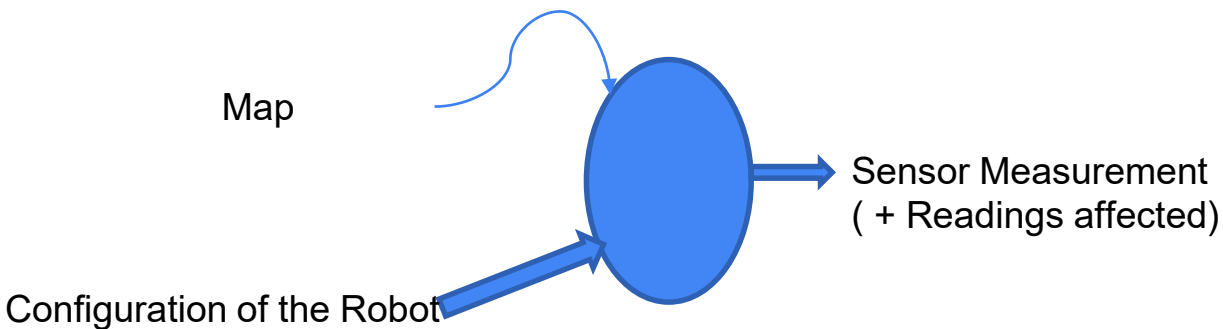
1:  Algorithm Particle_filter( $\mathcal{X}_{t-1}, u_t, z_t$ ):
2:     $\bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$ 
3:    for  $m = 1$  to  $M$  do
4:      sample  $x_t^{[m]} \sim p(x_t \mid u_t, x_{t-1}^{[m]})$ 
5:       $w_t^{[m]} = p(z_t \mid x_t^{[m]})$ 
6:       $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[m]}, w_t^{[m]} \rangle$ 
7:    endfor
8:    for  $m = 1$  to  $M$  do
9:      draw  $i$  with probability  $\propto w_t^{[i]}$ 
10:     add  $x_t^{[i]}$  to  $\mathcal{X}_t$ 
11:    endfor
12:    return  $\mathcal{X}_t$ 

```

Implementation of this technique will be detailed via both mathematical problem as well as lab demo in next module live sessions

Table 4.3 The particle filter algorithm, a variant of the Bayes filter based on importance sampling.

Introduction to Sensors & Sensor Model



Recap

1. Initial Belief
2. Control Update
3. Measurement Update

X_{t-1}	U_t	$P(X_t = \text{is_open})$	$P(X_t = \text{is_closed})$
is_open	push	1	0
is_closed	Push	0.8	0.2
is_open	do_nothing	1	0
is_closed	do_nothing	0	1

SENSOR MEASUREMENT MODEL : $P(Z_t | X_t)$

X_t	$P(Z_t = \text{sense_open})$	$P(Z_t = \text{sense_closed})$
is_open	0.6	0.4
is_closed	0.2	0.8

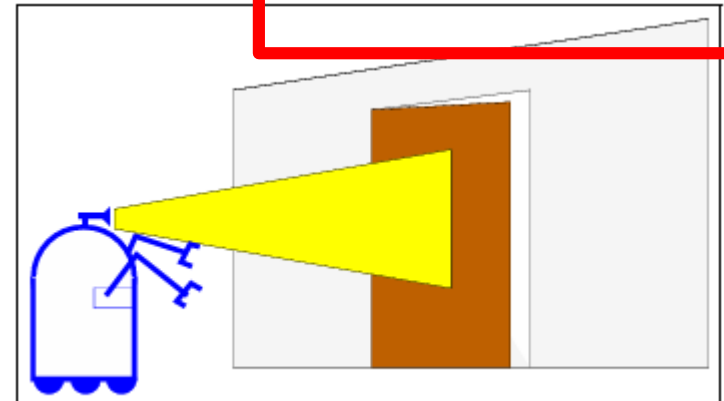
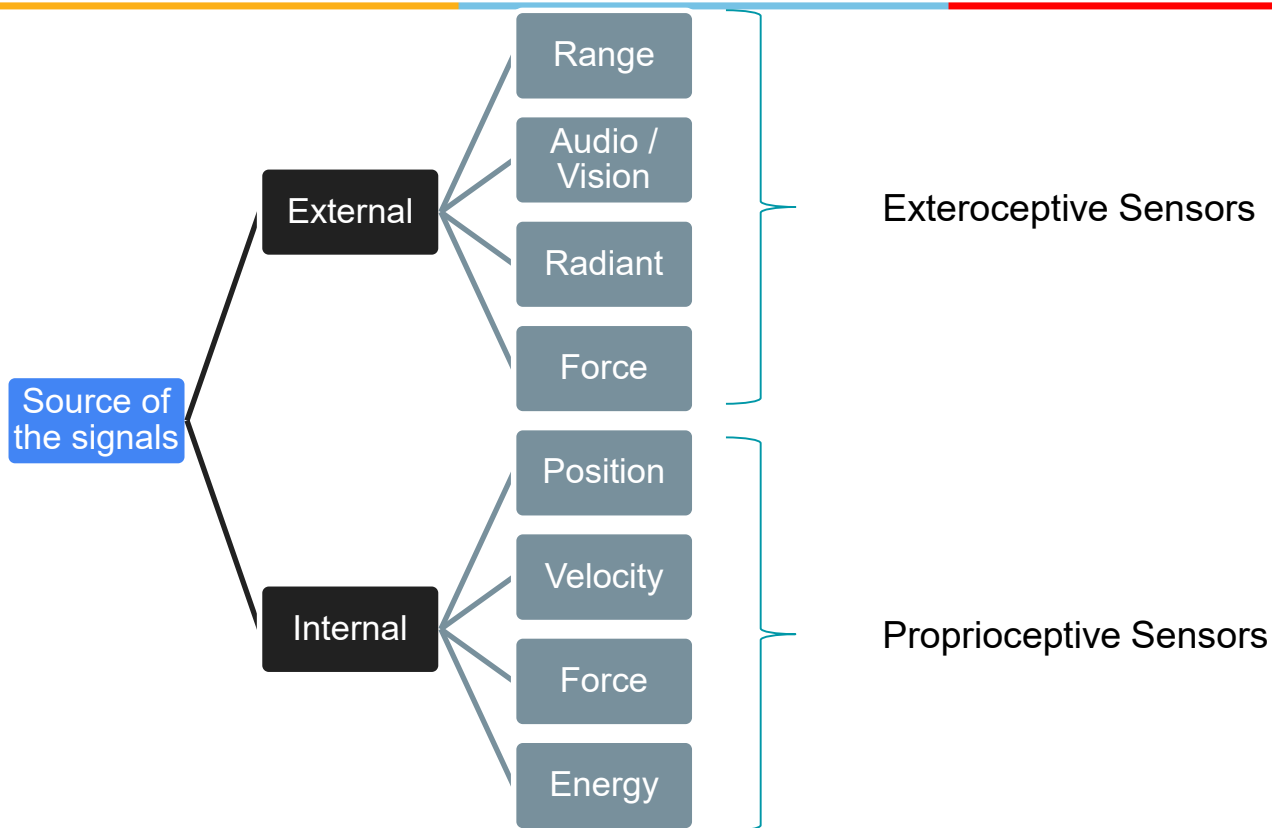
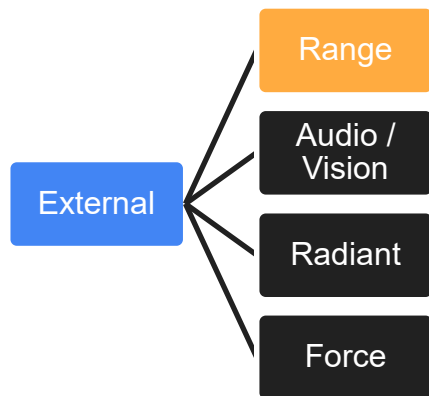


Figure 2.2 A mobile robot estimating the state of a door.

Types of Sensors



Exteroceptive Sensors



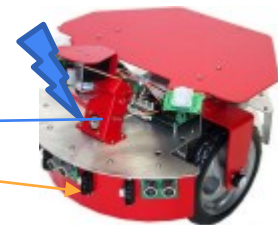
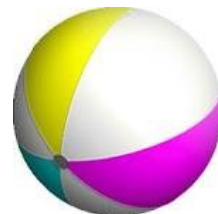
Triangulation

TOF: Time-of-Flight

Structured Light

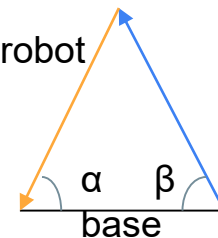
Phase Shift

Stereo Vision

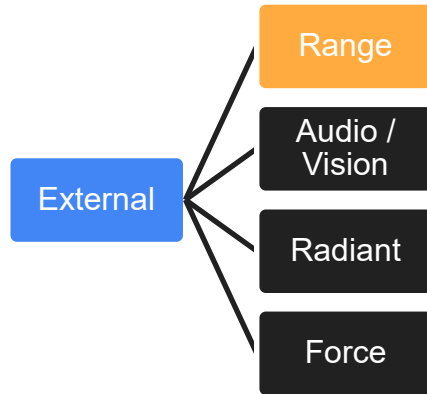


Distance of ball from robot

$$= \frac{base * \sin \alpha * \sin \beta}{\sin(\alpha + \beta)}$$
$$= \frac{base}{\tan \beta}$$



Exteroceptive Sensors



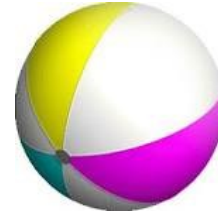
Triangulation

TOF: Time-of-Flight

Structured Light

Phase Shift

Stereo Vision



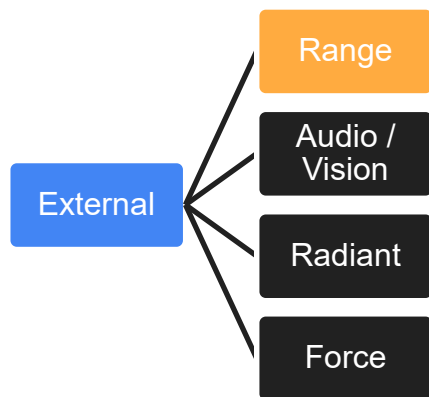
$$\text{Distance of ball from robot} = \frac{3 * 10^8 * 10 * 10^{-9}}{2} = 1.5 \text{ meter}$$

LiDAR
Radar

$$\text{Distance of ball from robot} = \frac{343 \frac{m}{s} * 6 * 10^{-3}}{2} = 1.03 \text{ meters}$$

Ultrasonic

Exteroceptive Sensors



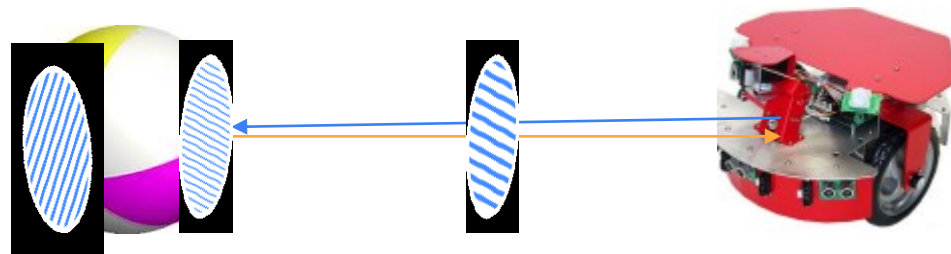
Triangulation

TOF: Time-of-Flight

Structured Light

Phase Shift

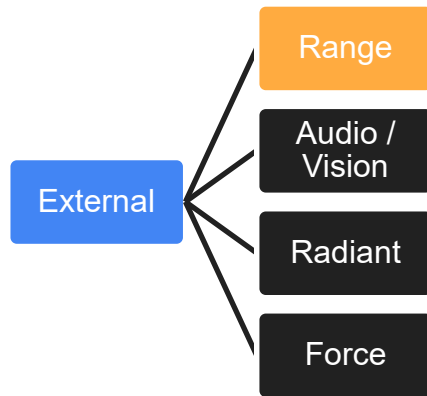
Stereo Vision



Distance of ball from robot
= 5 pixel displaced
= equivalent depth in meter



Exteroceptive Sensors



Triangulation

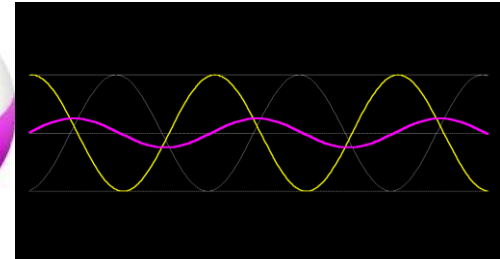
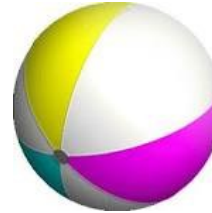
TOF: Time-of-Flight

Structured Light

Phase Shift

Photonic Mixer Device
(PMD)

Stereo Vision

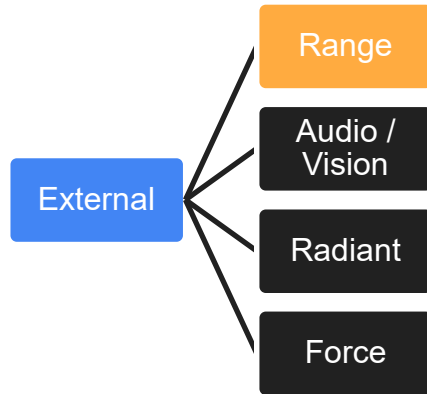


$$\begin{aligned}\text{Distance of ball from robot} &= \lambda \cdot \phi / 2\pi \\ &= (2\text{meter} * \pi/2 \text{radian}) / 2\pi \\ &= 0.5 \text{ meters}\end{aligned}$$

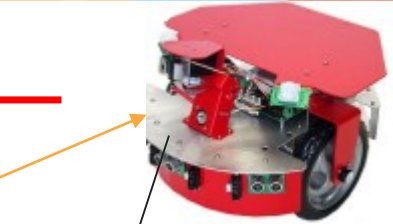
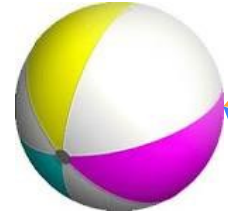
<https://youtu.be/WhBwMqcPo-A>

PMD is not same as Phase Shift . They overlap in few of the underlying principle

Exteroceptive Sensors



Triangulation
TOF: Time-of-Flight
Structured Light
Phase Shift
Stereo Vision



More on Perception & Sensor model will be discussed by Prof. Sangeetha in upcoming sessions

$$\begin{aligned} &\text{Distance of ball from robot} \\ &= \frac{\text{focal length of camera} * \text{Baseline Distance}}{\text{Difference in the obj appearance to camera 1 vs 2}} \\ &= \frac{800 \text{ pixel} \& 0.1 \text{ meter}}{10} \\ &= 8 \text{ meters} \end{aligned}$$

Required Reading

- T1 Chapter #5, #6

T2 Chapter #5, #6.1, #6.2

Next Class Plan

- ✓ Introduction to Motion Models (to be continued. Only brief Lab demo was done in this class)
- ✓ Sensor Measurement Models
- ✓ For the Next Class Prerequisite (Refer to the eLearning portal)